

SMART INDIA HACKATHON 2026

TITLE PAGE

• Problem Statement ID	SIH26236
• Problem Statement Title	AI-Based Intelligent Food Packaging Material Recommendation System for Food Commodities
• Theme	Agriculture, FoodTech & Rural Development
• PS Category	Software
• Team ID	161280
• Team Name (Registered on portal)	Evergreen

PACKWISE AI — Physics-informed hybrid AI platform matching 100+ Indian food commodities & storage physics with 31 barrier polymers.
✓ Explainable • ✓ Source-traced • ✓ Respiration & MAP aware • ✓ Edge AI Vision & Voice

100+ Indian Commodities DB (FDC+ICAR)	Mass-Transfer & Respiration Physics	Dynamic Storage & Transit Logistics	31 Polymers & Bio-Resins Catalog
TOPSIS Multi-Criteria Ranking	Dynamic Barrier Needs (OTR / WVTR)	MAP Equilibrium Solver (O2 / CO2)	Explainable Report, FSSAI Pass & QR

- ❖ How it addresses the problem:
- **Perishable crop devastation:** India suffers ~30% post-harvest spoilage (■1.52 Lakh Cr loss) due to unvented polybags and condensation rot.
 - **One pack can't fit all:** Moisture sorption, fat oxidation, and respiration rates differ fundamentally across commodities; mismatched films cause premature spoilage.
 - **Selection is expert-dependent & slow:** Fragmented offline knowledge, arbitrary polymer thickness, and high trial-and-error laboratory costs.
 - **Ground-level stakeholders struggle:** Smallholder farmers, FPOs & food MSMEs lack packaging scientists and need instant guidance in native language.
 - **Unified physics + regulatory decision core:** Integrates food composition, crop biology, polymer barrier physics, and FSSAI/BIS rules into one automated engine.

SOLUTION PIPELINE: Food Selection (100+ Crops) → Property Analysis (aw, pH, Lipids) → Risk Analysis (Respiration & O2) → Compatibility (31 Polymers) → TOPSIS AI Ranking → Explained Result (3D Lab + QR)

1. EXPLAINABLE AI Transparent mathematical justification (TOPSIS Closeness Ci*) with statutory FSSAI citations & Gemini Flash reasoning.	2. MASS-TRANSFER PHYSICS Tetens vapor pressure equation + Arrhenius respiration kinetics for anaerobic hazard alerts.	3. 3D PACKAGING LAB & HUD Interactive Three.js WebGL multi-layer digital twin with layer exploding + Quantum 3D cards & 50-frame storyboard.	4. SOURCE TRACEABILITY Strict data discipline: FSSAI, APEDA, BIS, ICAR & USDA FDC — zero fabricated synthetic values.	5. AASAAN VOICE + AI VISION Bilingual Hindi/English voice assistant (Vaani AI) + Real-time Edge YOLOv8 & NVIDIA NIM camera crop scanner + batch QR pass.
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Evergreen Team 161280		TECHNICAL APPROACH & ARCHITECTURE AI-Based Intelligent Food Packaging Material Recommendation System • PS SIH26236		SMART INDIA HACKATHON 2026	
FRONTEND TECH STACK • React 19, Vite (Instant HMR) • TypeScript, Tailwind CSS • Three.js WebGL 3D Film Lab • Quantum 3D Perspective HUD • 50-Frame Visual Storyboard • Web Speech Voice AI & WebRTC		BACKEND & DATABASE • Python 3.12 High-Speed Async • FastAPI (RESTful, Auto-Docs) • Uvicorn ASGI, Pydantic v2 • SQLAlchemy ORM • SQLite Offline / PostgreSQL Cloud • Resilient Client-Side Engine		AI & MACHINE LEARNING (NEW) • YOLOv8n Edge Neural Net: 12ms/60FPS local on-device scanner • NVIDIA NIM Llama-3.2: Multimodal GPU vision detector • Gemini 3.8/1.5 Flash: FSSAI explainability & reasoning • NumPy TOPSIS: MCDM decision • Vaani Voice AI: Hindi/Eng STT/TTS	PHYSICS & KINETIC MODELS • Tetens Formula: Saturated vapor pressure $P_{sat}(T)$ & WVTR flux • Michaelis-Menten: Dynamic O_2/CO_2 respiration kinetics • Arrhenius Model: Temperature buffer (+5°C safety) • MAP Solver: Laser micro-perforation gas balance
OPERATIONAL METHODOLOGY PIPELINE: 1. Crop Intake & Profiling: Input via Hindi Voice or Camera (YOLOv8/NVIDIA NIM); extract moisture %, aw, pH, lipids & respiration. 2. Physics & Respiration Modeling: Tetens equation calculates $P_{sat}(T)$; Michaelis-Menten & Arrhenius model O_2/CO_2 with +5°C safety buffer. 3. Barrier Solving: Computes target OTR & WVTR bounds; filters 31 food-grade barrier polymers & bio-resins. 4. TOPSIS MCDM: Vector-normalized ranking balances Shelf-Life Extension (Max), Film Cost (Min), and Carbon Footprint (Min). 5. Regulatory Audit: Verifies FSSAI 2018 (IS 9845) migration compliance; renders Three.js 3D pack and batch QR pass.				INTELLIGENT MULTI-LAYER FOOD PACKAGING ARCHITECTURE: • Outer Protective Layer: BOPET (12 μm) → Mechanical puncture resistance & printable surface • Adhesive Tie Layer: Modified PE (3 μm) → Inter-layer molecular bonding preventing delamination • High Gas Barrier: EVOH Core (5 μm) → Ultra-low OTR (<1 cc/m²·day) • Prevents oxidation & rancidity • Adhesive Tie Layer: Modified PE (3 μm) → Inter-layer molecular bonding preventing delamination • Hermetic Sealant: Metallocene LLDPE (50 μm) → Moisture barrier (WVTR <3 g/m²·day) • FSSAI Food Contact Safe (IS 10146) • MAP Gas Equilibrium: Laser micro-perforations match respiration (RO₂) to extend produce shelf life by 200% - 300%.	
SYSTEM ARCHITECTURE FLOW (END-TO-END MULTI-TIER CONNECTION): 1. Client Layer: React 19 UI, Quantum 3D HUD, Vaani Voice AI, WebRTC YOLOv8/NVIDIA Scanner 2. API Gateway: FastAPI Async Router, Pydantic v2 3. AI & Physics: YOLOv8 ONNX, NVIDIA NIM Vision, NumPy TOPSIS, Tetens/Arrhenius Solver 4. Data Layer: 100 Indian Crops (ICAR/USDA), 31 Polymers DB, FSSAI IS 9845 5. Delivery HUD: Three.js 3D Lab, MAP Solver, QR Pass					

Evergreen Team 161280		FEASIBILITY AND VIABILITY AI-Based Intelligent Food Packaging Material Recommendation System • PS SIH26236		SMART INDIA HACKATHON 2026
TECHNICAL FEASIBILITY ✓ Modern full stack — React 19 + FastAPI ✓ Sub-50ms execution latency for TOPSIS & physics ✓ 12ms / 60 FPS real-time YOLOv8 edge vision ✓ Modular, high-speed RESTful microservices ✓ Three.js WebGL & Quantum 3D run in any browser ■ 100% functional working prototype verified locally & deployed live.		DATA RIGOR & PROVENANCE ✓ 100 pre-calibrated commodities across 6 core food categories ✓ Integrated official FSSAI • APEDA • BIS • ICAR-CIPHET standards ✓ Complete provenance lineage kept for every value & equation ✓ Strict zero synthetic/hallucinated data — fully explainable math ■ Every rule, threshold & barrier value traceable to Slide-6 sources.		DEPLOYMENT & SCALING ✓ Fully functional cloud & edge deployable prototype with offline mode ✓ Zero proprietary hardware Capex needed (any smartphone or PC) ✓ Stateless FastAPI containers scale horizontally with traffic load ✓ Direct integration pathway with e-NAM digital mandis & APEDA ■ Built for immediate grassroots deployment across rural FPOs.
❖ Potential challenges and risks ➔ Strategies for overcoming these challenges				
1. Crop respiration & moisture vary by harvest maturity ➔ Calibrated against ICAR & UC Davis postharvest baselines; live sliders allow fine-tuning.				
2. Broken Indian rural cold-chain transit abuse ➔ Arrhenius kinetic engine factors in +5°C transit safety buffer to prevent premature spoilage.				
3. Material barrier variations across film suppliers ➔ Normalized ASTM/ISO testing metrics (OTR: ASTM D3985, WVTR: ASTM F1249) stored in DB.				
4. Complex Plastic Waste Management (PWM) Rules 2022 ➔ Automated regulatory audit checks FSSAI (IS 9845) migration and prioritizes compostable PLA/PHA.				
5. Rural digital literacy divide for grassroots farmers ➔ Aasaan Mode with native Hindi voice assistance (Vaani AI) and real-time camera crop recognition.				
✓ Verdict: Fully feasible with modern web & physics AI stack — validated working prototype ready for immediate pilot. • Risk: Low–Medium				

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Team 161280

IMPACT AND BENEFITS

SMART INDIA
HACKATHON 2026

AI-Based Intelligent Food Packaging Material Recommendation System • PS SIH26236

TARGET AUDIENCE: Farmers / Mandi FPOs | Food Processors | Agri-MSMEs | Packaging Designers | Exporters | Cold-Chain Logistics | Food Scientists | Govt Inspectors

SOCIAL • Democratizes packaging science for grassroots farmers • Native Hindi/English voice AI bridges literacy barrier • Prevents distress sales during market glut via shelf-life gain • Protects smallholder farm incomes & rural livelihoods	ECONOMIC • Reduces post-harvest spoilage by 25% – 40% across transit • Mitigates India's annual █1.52 Lakh Crore food loss burden • Saves 12-20% resin cost via precision film down-gauging • Eliminates costly export consignment rejections at ports	AGRICULTURAL • Crop-wise preservation for 100 Indian commodities • Preserves fresh produce vitamins, firmness & natural aroma • Enables long-distance rail/road freight without spoilage • Generates export-ready APEDA packaging compliance passes	ENVIRONMENT • Recommends compostable Bio-PLA, Bio-PBS & PHA films • Accelerates transition to recyclable mono-material polyolefins • Fully aligns with Plastic Waste Management (PWM) Rules 2022 • Minimizes methane emissions from rotting mandi landfill waste	FOOD QUALITY • Optimal OTR/WVTR stops moisture pooling & fungal rot • MAP gas equilibrium (O2/CO2) preserves crispness & color • Prevents lipid oxidation and rancidity in high-fat foods • Guarantees certified, safe & hygienic produce for consumers
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ADOPTION PATHWAY: 1. PROTOTYPE (Functional Full-Stack) → 2. PILOT (Selected Mandis & FPOs) → 3. VALIDATION (IIP & Lab Trials) → 4. SCALE-UP (National e-NAM / Cloud)

Compare Recyclable & Compostable Options	Right-Size & Down-Gauge Packs (Less Plastic)	Voice-First, Low-Literacy Bilingual UI	Rural + Mandi FPO Friendly Practical Guidance
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FUTURE SCOPE & ADVANCED EXTENSIONS:

- 1. National e-NAM Mandi Integration:** Direct API integration with electronic mandis for automated packaging grading and instant transit passes.
- 2. IoT Telemetry & Cold-Chain Sensors:** Live RFID/BLE temperature & RH data streams for real-time dynamic shelf-life decay updates.
- 3. Active & Antimicrobial Bio-Films:** Integration of natural nano-emulsion coatings (chitosan, thymol) to actively arrest fungal decay.
- 4. Converter Marketplace & Procurement:** Direct B2B digital gateway connecting FPOs directly with verified sustainable film converters.
- 5. Offline Progressive Web App (PWA):** Zero-connectivity Service Worker caching for seamless use in remote rural mandis.
- 6. AI Computer Vision Ripeness Grading:** Camera-based maturity & blemish detection prior to barrier film recommendation.

OFFICIAL GOVERNMENT & SCIENTIFIC SOURCES (VERIFIED PORTALS):

- **FSSAI [fssai.gov.in]:** Packaging Reg. 2018 & Overall Migration Limits (IS 9845:2020)
- **BIS [bis.gov.in]:** Indian Standards: IS 10146 (PE food contact), IS 12252 (PET), IS 15495
- **APEDA [apeda.gov.in]:** Fresh produce export packaging standards & cold-chain phytosanitary norms
- **IIP [iip-in.com]:** Indian Institute of Packaging (ASTM D3985 OTR & ASTM F1249 WVTR)
- **ICAR-CIPHET [ciphnet.icar.gov.in]:** Post-harvest loss statistics & crop respiration rates (RO2/RCO2)
- **MoFPI [mofpi.gov.in]:** Ministry of Food Processing Industries cold-chain cluster schemes
- **USDA FoodData [fdc.nal.usda.gov]:** Crop baseline composition dataset (moisture %, Aw, lipids, pH)
- **CPCB (MoEFCC) [cpcb.nic.in]:** Plastic Waste Management Rules 2022 & Compostable IS/ISO 17088

EVIDENCE DISCIPLINE — ZERO FABRICATED SYNTHETIC VALUES: Official / Verified • Authoritative Research • Derived Physics • Statutory Compliance

PROTOTYPE DEMO PLAN — WHAT WE WILL DEMONSTRATE IN EVALUATION (TESTED & WORKING):

1. Live 100-Crop Search & Camera Scan
Real-time edge YOLOv8 + NVIDIA NIM
Vision crop scanner with auto-detection.

2. Mass-Transfer & Respiration Solver
Live Tetens equation vapor pressure &
Arrhenius Michaelis-Menten respiration flux.

3. Quantum 3D TOPSIS & Three.js Lab
Interactive 3D multi-layer film digital twin
with layer exploding & MCDM sliders.

4. Bilingual Voice AI & FSSAI QR Pass
Vaani Hindi/English voice copilot, Farmer
Aasaan mode & instant batch QR report.